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MILITARY SPECIFICATION

SERVICE UNIT, FLAME THROWER, TRUCK MOUNTED, M4A2

1. SCOPE

1.1 This specification covers a service unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date for invitation for bids or request far proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

P-D-680	-	Dry Cleaning Solvent.
NN-P-515	-	Plywood, Container Grade.
TT-C-520	-	Coating, Underbody (For Motor Vehicles).
TT-E-527	-	Enamel, Alkyd, Lusterless-.
TT-P-664	-	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
TT-T-291	-	Thinner; Paint, Volatile Mineral Spirits (Petroleum-Spirits).
TT-V-51	-	Varnish: Asphalt.
UU-T-81	-	Togs, Shipping and Stock.
VV-F-800	-	Fuel Oil, Diesel.
VV-G-76	-	Gasoline, Automotive.
VV-L-800	-	Lubricating Oil, General Purpose, Preservative, (Water-Displacing, Low Temperature).

FSC 1040

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PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-T-60 - Tape; Pressure Sensitive Adhesive, Waterproof-For Packaging and Sealing.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-130 - Paper, Wrapping, Laminated and Creped.
MIL-R-196 - Repair Parts far Internal Combustion Engines, Packaging Of.
MIL-S-207 - Sulfuric Acid, Electrolyte, Packaging, Packing, and Marking far Shipment and Storage Of.
MIL-P-208 - Packaging and Packing for Shipment and 9torage of Batteries, Storage, Automotive and Navy Portable.
MIL-C-450 - Coating-Campaund, Bituminous Solvent Type, Black (For Ammunition).
MIL-T-704 - Treatment and Painting of Material.
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose.
MIL-P-3542 - Primer: Pressure-Sensitive Tape.
MIL-H-5606 - Hyadraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordinance.
MIL-V-13811 - Varnish, Waterproofing, Electrical, Ignltion.
MIL-L-15017 - Lubricating Oil, Hydraulic.
MIL-H-15424 - Hand Tools: Packaging Of.
MIL-C-17504 - Coating Compound, Acrylic, Clear.
MIL-C-51172 - Compressor, Reciprocating, 50 CFM, 3000 PSI.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging Of.

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DRAWINGS

EDGEWOOD ARSENAL

DL81-8-751 - Service unit, Flame Thrower, Truck-Mounted, M4A2-D.
DL81-8-1049 - Service Unit, Flame Thrower, Truck-Mounted, M4A2-IR .

(All drawings referenced in this specification are listed on Dwg DL81-8-751 or DL81-8-1049.)

(Copies of specifications, standards, drawings and publicatlons required by suppliers in connection with specific procurerment functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated. the issue in effect on date of invitation for bids or request for Proposal shall apply.

THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Displacement Compressors, Vacuum Pumps and Blowers.

(Copies may be obtained from the American Society of Mechanical Engineers, 345 East 47th Street, New York 17, New York.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL81-8-751 or DL81-8-1049 or in subsidiary drawings shall confcrm to the specifications listed thereon, or to the specific characteristics set forth in the drawings.

3.1.2 Components. All components of the service unit shalll conform to the specifications and drawings listed on Drawing DL81-8-751 or DL81-8-1049 and subsidiary drawings.

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3.2 Truck modification. Modifications to the M35A1 truck shall conform to Drawing E81-8-940 or E81-8-1053.

3.3 Assembly. The service unit shall be assembled in accordance with Drawing E81-8-751 or E81-8-1049. The hydraulic piping assembly (E81-1-991) shall be filled with one of the following:

(a) For service between 32° and 125°F, hydraulic oil conforming to Military Symbol 2135H of MIL-L-15017.

(b) For service below 32°F, hydraulic oil conforming to MIL-H-5606.

3.4 Performance.

3.4.1 Compressor preheating. The engine coolant when at operating temperature of 160° to 180°F shall heat the compressor preheating system when tested as specified in 4.6.1.

3.4.2 Fuel heating. The engine coolant (3.4.1) shall heat the heater jacket and mixing tank heating system when tested as specified in 4.6.1.

3.4.3 Hydraulic pump.

3.4.3.1 Fan motor. The hydraulic pump shall drive the fan motor, for compressor preheating, at a speed of no less than 1750 revolutions per minute (rpm) when tested as specified in 4.6.1.

3.4.3.2 Fuel pump.

3.4.3.2.1 Intake. The hydraulic pump shall drive the fuel pump to supply a minimum of 200 gallons of oil conforming to VV-L-800 and have a minimum suction head of 10 feet. The oil shall be at a temperature of 75° ± 5°F. The fuel pump shall pump the 200 gallons of oil into the mixing tank in a maximum time of 7 minutes when tested as specified in 4.6.1.

3.4.3.2.2 Discharge. The fuel pump shall pump a minimum of 200 gallons (see 3.4.3.2.1) out of the mixing tank in a maximum time of 13 minutes against a 125 pound per square inch gage (psig) minimum back pressure when tested as specified in 4.6.1.

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3.4.3.3 Agitator motor. The hydraulic pump shall drive the agitator motor when tested as. specified in 4.6.1.

3.4.4 Compressor air system.

3.4.4.1 compressor shaft speed The truck engine governor linkage shall control the engine speed at 1800 ± 50 rpm so that the truck power take-off engaged, the transmission in fourth gear, and the transfer case in neutral, the compressor shall be driven at its rated speed when tested as specified in 4.6.1.

3.4.4.2 Performance.

3.4.4.2.1 Three thousand psig air system. The 3000 psig air system shall furnish air to its hose connection at a presure of 3000 ± 200 psig and at a minimum rate of 50 cubic feet per minute (cfm) when tested as specified in 4.6.1.

3.4.4.2.2 Two thousand psig air system. The 2000 psig air system shall furnish air to its hose connection at a pressure of 2000 ± 150 psig when tested as specified in 4.6.1.

3.4.5 Leakage. The fuel mixing, hydraulic, coolant and air system shall not leak when tested as specified in 4.6.2.

3.5 Suction pipe strainer. The strainer shall meet the requirements of Drawing B81-8-978 when tested as specified in 4.6.3.1.

3.6 Heat exchanger. The heat exchanger shall meet the requirements of Drawing 581-8-847 and not leak when tested as specified in 4.6.3.2.

3.7 Compressor heat radiator. The radiator shall meet the requirements of Drawing D81-8-1028 and not leak when tested as specified in 4.6.3.3.

3.8 Check valve. The valve shall meet the rquirerments of Drawing B81-8-898 when tested as specified in 4.6.3.4.

3.9 Safety relief valves.

3.9.1 Functioning. The valves shall meet the requirements of Drawing B81-8-897 when tested as specified in 4.6.3.5.1 and 4.6.3.5.4.

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3.9.2 Leakage. The valves shall not leak when tested as specified in 4.6.3.5.2 and 4.6.3.5.5.

3.9.3 Proof pressure. The valves shall not fail under their associated proof pressures when tested as specified in 4.6.3.5.3 and 4.6.3.5.6.

3.10 Needle valve. The valve shall meet the requirements of Drawing B81-8-927 and shall not leak when tested as specified in 4.6.3.6.

3.11 Fuel nozzle valve.

3.11.1 Functioning. The valve shall open with firm hand pressure and meet the requirements of Drawing C81-8-840 when tested as specified in 4.6.3.7.1.

3.11.2 Leakage. The valve shall not leak when tested as specified in 4.6.3.7.2.

3.12 Fuel valves. The valve control, Drawing C81-8-827-1, shall operate with a torque of 55 ± 5 foot pounds. The valve control, Drawing C81-8-827-2, shall operate with a torque of 30 ± 5 foot pounds. Both valves shall not leak internally or externally at a hydrostatic pressure of 200 ± 10 psig when tested as specified in 4.6.3.8.

3.13 Hose assemblies. The hose assemblies shall meet the requirements in Drawings C81-8-915, B81-8-954, C81-8-977, and C81-981 and shall not leak when tested as specified in 4.6.3.9.

3.14 Line filter. The filter shall meet the requirements of Drawing C81-8-994 and shall not leak when tested as specified in 4.6.3.10.

3.15 Air receiver. The air receiver shall meet the requirements of Drawing C81-8-893 when tested as specified in 4.6.3.11.

3.16 Hydraulic tank (Dwg D81-8-831). The tank shall not leak at an air pressure of 5 psig when tested as specified in 4.6.3.12.

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3.17 Hydraulic motors.

3.17.1 Functioning. The motors shall meet the requirements of Drawings C81-8-845, C81-8-846, and C81-8-912 when tested specified in 4.6.3.13.1.

3.17.2 Leakage. The motors shall not leak when tested as specified in 4.6.3.13.2

3.18 Hydraulic pump.

3.18.1 Functioning. The pump shall meet the requirements of Drawing C81-8-844 when tested as specified in 4.6.3.14.1.

3.18.2 Leakage. The pump shall not leak when tested as specified in 4.6.3.14.2.

3.19 Fuel pump.

3.19.1 Functioning. The pump shall meet the requirements of Drawing D81-8-843 when tested as specified in 4.6.3.15.1 and 4.6.3.15.2.

3.19.2 Leakage. The pump shell not leak under a pressure of 175 psig when tested as specified in 4.6.3.15.3.

3.20 Treatment and pointing. All parts of the service unit that would normally be painted In commercial practice shall be treated and painted in accordance with information within the specifications specified on the drawings listed in Drawing DL81-8-751 and DL81-8-1049.

3.21 Identification marking. Each service unit shall be serially numbered. Additional identification plates required by using services shall be as specified (see 6.2).

3.22 Stake. The static ground stake (Dwg C81-8-936) shall meet the torque requirement in the drawing when tested as specified in 4.6.3.16.

3.23 Workmanship. The service unit shall be free from damage such as dents, cracks or deformation which could affect its intended use.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specificat ion where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, 3.3, 3.20, 3.21, 3.23 and section 5 for which specific inspection has nor been provided in this specification have been satisfied.

4.3 Component lotting. A lot shall consist of like components, produced by one manufacturer, from the some materials, under essentially the same manufacturing conditions.

4.4 Sampling of components for examination and tests. Sampling shall be conducted in accordance with MIL-STD-105. For proof pressure level S-2 shall be used.

4.5 Inspection provisions.

4.5.1 Component examination and nondestructive tests. Sample components and the level A and level B preparation for delivery thereof shall be examined for all the requirements of the drawings and this specification and the classification of defects as applicable and tested in accordance with 4.6.3 and MIL-STD-105. The AQL far components shall be 0.25 percent defective for acceptance. Proof pressure shall use an AQL of 2.5 percent defective far acceptance.

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4.5.2 Service unit. Each service unit shall be examined and tested in accordance with 4.6.1 and 4.6.2.

4.5.3 Classification of defects.

4.5.3.1 Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Tags not processed as specified
102	Preservatives or coatings missing
103	Openings, covers, and doors not processed as specified
104	Equipment not lubricated as specified
105	Item improperly secured to prevent movement
106	Strapping loose or missing
107	Marking missing, illegible, or incorrect

4.6 Tests.

4.6.1 Performance. In the following specified order, the service unit shall be tested at any ambient temperature, for compliance with 3.4:

(a) Start truck engine and allow to warm-up to 160°F. Open preheat valve to allow coolant to circulate through the compressor preheating system.

(b) Close both valves (compressor preheat and heat exchanger) and operate fan motor for compressor preheating. Record the fan shaft speed using a stroboscope or other acceptable method; then stop fan.

(c) Pump oil (see 3.4.3.2.1) into mixing tank. Record the number of gallons pumped, pumping time, suction head and temperature of test fluid.

(d) After tank is filled in accordance with 3.4.3.2.1, start the agitator motor and check through top of tank to ascertain whether the agitator is functioning. Then stop agitator.

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(e) Pump oil (see 3.4.3.2.2) out of mixing tank. Record number of gallons pumped, pumping the, back pressure and temperature of test fluid.

(f) Start compressor and record the shaft speed using a tachometer.

(g) With compressor running, record the cfm in accordance with ASME Code for Displacement Compressors, Vacuum Pumps and Blowers and the psig of both hose connections.

4.6.2 Leakage.

4.6.2.1 Fuel system. service unit operating, check all piping, joints, and fittings of t e fuel system for visible signs of leakage with the controls set at the following operating positions:

(a) Filling mixing tank from oil (see 3.4.3.2.1) supply.

(b) Heating test fluid.

(c) Discharging test fluid from mixing tank.

4.6.2.2 Hydraulic piping. With the service unit operating, check the hydraulic lines, joints and fittings for visible signs of leakage during the following operations:

(a) Compressor preheating.

(b) Agitation.

(c) Discharging test fluid from mixing tank.

4.6.2.3 Coolant system. With the service unit operating and coolant control valves open, check all piping, ioints, and fittings of the coolant system for visible signs of leakage.

4.6.2.4 Air piping. Both air service outlet valves shall be closed and the compressor operated to pressurize the 3000 psi air lines to 3000 ± 200 psig. All pressurized air line ioints shall be checked for leakage with a soap solution.

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4.6.3 Component. The components shall be tested as indicated below. Each component be tagged or marked upon the successful completion of test.

4.6.3.1 Suction pipe strainer. Connect the strainer to a controlled pressure source of gasoline conforming to VV-G-76 and record gasoline temperature, inlet and outlet pressures, and flow rate.

4.6.3.2 Heat exchanger.

4.6.3.2.1 Functioning. Connect the tube side of the exchanger to a suitable oil source and the shell side to a suitable water circulating system and function the exchanger for compliance with 3.6.

4.6.3.2.2 Leakage.

4.6.3.2.2.1 Shell side. Provide a suitable plug at the outlet of the shell side of heat exchanger. Hydrostatically test at test pressure for visible indication of leakage.

4.6.3.2.2.2 Tube side. Provide a suitable plug at the outlet of the tube. side of heat exchanger. Hydrostatically test at test pressure for visible indication of leakage.

4.6.3.3 Compressor heat radiator. Plug the radiator outlet and connect the inlet to a hydrostatic pressure. Hold the pressure far 2 minutes and check the radiator for working pressure and visible leakage.

4.6.3.4 Check valve. Connect the valve to a pneumatic test setup and record the test pressure, cracking pressure, pressure drop at full air flow rote, operating pressure, back pressure, and leakage in cubic inches per minute.

4.6.3.5 Safety relief valves.

4.6.3.5.1 Functioning of B81-8-897-1. The valve shall be connected to an air pressure source control lable from 0 to 3000 psig. increase pressure 50 psig at a time from 2000 psig until a pressure of 2300 psig has been reached. The pressure at which the valve starts to open and the flow rate at 2300 psig shall be recorded.

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4.6.3.5.2 Leakage of B81-8-897-1. With the valve connected as specified in 4.6.3.5.1 submerge the valve in water and increase pressure 50 psig at a time unti a pressure of 2000 psig has been reached. Record the pressure at which any leakage is indicated.

4.6.3.5.3 Proof pressure of B81-8-897-1. The valve shall be connected to an air pressure source controllable from 0 to 4000 psig, and sujected to a proof pressure of 3450 psig.

4.6.3.5.4 Functioning of B81-8-897-2. The valve shall be connected to an air pressure source controllable from 0 to 4000 psig. Increase pressure 50 psig at a time from 3000 psig until a pressure of 3300 psig has been reached. The pressure at which the valve starts to open and the flow rate at 3300 psig shall be recorded.

4.6.3.5.5 Leakage of 881-8-897-2. With the valve connected as specified in 4.6.3.5.4 submerge the valve in water and increase pressure 50 psig at time until a pressure of 3000 psig has been reached. Record the pressure at which any leakage is indicated.

4.6.3.5.6 Proof pressure of B81-8-897-2. The valve shall be connected to an air pressure source controllable from 0 to 5000 psig, and subjected to a proof pressure of 4950 psig.

4.6.3.6 Needle valve. Connect the valve to a hydraulic test setup. Close handle by hand and hold at operating pressure for 2 minutes. Record temperature of water in degrees Fahrenheit, and note any leakage.

4.6.3.7 Fuel nozzle valve.

4.6.3.7.1 Functioning. Connect the inlet of the valve to a water pressure source. Note ease with which the valve opens and closes and record flow rote, inlet pressure and water temperature.

4.6.3.7.2 Leakage. Connect the valve as specified in 4.6.3.7.1. With the valve closed, note any indication of leakage when valve is subjected to an inlet pressure of 125 psig for 2 minutes.

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4.6.3.8 Fuel valves.

4.6.3.8.1 Ease of operation. Operate each sample valve by hand and note ease of operation when pressurized.

4.6.3.8.2 Leakage. Connect each sample valve to a 0 to 200 psig hydrostatic source and check all sealed joints far any leakage.

4.6.3.9 Hose assemblies. Plug one end of the hose with a suitable adapter and connect the other end to a hydrostatic pressure source. Apply the required maximum operating pressure and let the hose assembly remain pressurized for 5 minutes.

4.6.3.10 Line filter. Connect the filter to a controlled pressure source of oil (see 3.4.3.2.1). Record inlet and outlet pressure, flow rate and any visible evidence of leakage at the operating inlet pressure.

4.6.3.11 Air receiver. The permanent volumetric expansion of the air receiver shall be determined by water-jacket or other method suitable for obtaining accurate data. Subject the receiver to an internal water pressure of 3500 ± 50 psig for a period of 1 minute (see 6.4 and 6.5). The pressure gage shall be accurate to within 1 percent or 0.1 cubic centimeter. Thoroughly dry receiver after test.

4.6.3.12 Hydraulic tank. Provide a suitable plug for each port and connect the tank to a pressure source. Hold tank at test pressure far 2 minutes using a soap solution to check for leaks.

4.6.3.13 Hydraulic motors.

4.6.3.13.1 Functioning. With the test equipment supplying hydraulic fluid and the motor operating at an ambient temperature of 75° ± 5°F check flow rate, revolutions per minute, horsepower delivery and pounds per square inch gage.

4.6.3.13.2 Leakage. Plug the outlet port of the motor and connect the inlet port to a 1500 + 50 psig hydraulic pressure source. Record any leakage.

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4.6.3.14 Hydraulic pump.

4.6.3.14.1 Functioning. With the test equipment supplying hydraulic fluid and motor operating at an ambient temperature of 750 ± 5°F check flow rate of hydraulic fluid, revolutions per minute, horsepower input, and pounds per square inch gage.

4.6.3.14.2 Leakage. Plug. the outlet port of the pump and connect the inlet port to o 1500 + 50 psig hydraulic pressure source. Record any leakage.

4.6.3.15. Fuel pump.

4.6.3.15.1 High discharge pressure. Operate the fuel pump at a high discharge pressure at an ambient temperature of 75° ± 5°F (see 3.4.3.2.2). Record the flow rate, discharge pressure, suction head, shaft speed, and power input.

4.6.3.15.2 Low discharge pressure. Operate the fuel pump at a low discharge pressure at an ambient temperature of 75° ± 5°F (see 3.4.3.2.2). Record the flow rate, discharge pressure, and suction head.

4.6.3.15.3 Leakage. Plug one port of the pump and test the pump hydrostatically. Observe the pump for leakage.

4.6.3.16 Stake. Test the static ground stake by holding the auger end securely and applying torque to the handle about the stake axis using a torque wrench and suitable adapter. Gradually increase torque (rotation direction optional) until the required value is reached or joint fails, whichever occurs first. Record the required value attained or value at which failure occurred.

5. PREPARATION FOR DELIVERY (For packaging of stocked repair parts, see 6.6.)

5.1 Disassembly and matchmaking.

5.1.1 Disassemble Unless otherwise specified in the contract or order, the extent of disassembly shall be confined to the removal of parts that would unnecessarily increase the cubage of the packed item, and shall be in accordance with the recommendations of the equipment manufacturer, subject to the approval

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of the procuring agency. All bolts, nuts, pins, and washers removed during disassembly shall be replaced on one of the mating ports and secured in such a manner as to prevent loss.

5.1.2 Matchmarking. The removed ports and their mating parts shall be identically marked by stenciling with either letters or numerals. Lusterless white enamel, conforming to TT-E-527 shall be used. Parts too small to accommodate stenciling shall have the matchmarking information indicated on either cloth shipping tags, conforming to type A of UU-T-81 or on metal tags and securely attached to the parts removed and the mating parts in such manner so as not to damage the item or the preservative thereon.

5.2 Cleaning and drying. After test runs, if any, have been completed, and before the application of any preservatives or coatings, all surfaces shall be cleaned and dried in conformance with MIL-P-116. Cleaning process C-1, and the most applicable of the drying procedures listed therein shall be used. In addition, special cleaning and drying methods covering certain areas on the equipment are indicated herein, and these methods whenever specified shall be used. Adequate precautions shall be taken to protect adjacent critical areas.

5.3 Repainting. Damaged portions of painted surfaces shall be treated and repainted in accordance with the applicable requirements of type A finishing system of MIL-T-704. The same quality and color of paint as specified on the applicable drawings shall be used.

5.4 Lubrication. After cleaning and repainting have been completed, all grease fittings shall be wiped clean. Joints, bearings, and similar moving parts shall be thoroughly lubricated with the equipment manufacturer's approved lubricants. Change of crankcase oil and servicing of enclosed gears is not included in this lubrication procedure. After lubrication, all grease fittings shall be wiped clean, and a red circle shall be painted around the base of each fitting. Grease fittings shall be covered with red plastic snap-on caps or with type III, class I, pressure-sensitive tape conforming to PPP-T-60.

5.5 Preservation and packaging. Levels shall be as specified (see 6.2).

5.5.1 Level A.

5.5.1.1 Preservatives and packaging methods. Unless otherwise specified, preservatives cited herein and the application thereof, shall conform to the applicable requirements of MIL-P-116. Preservatives shall not be applied to painted surfaces, and provisions shall be made to prevent such surfaces from

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coming in contact with preservatives on adjacent metal ports. Packaging methods cited herein shall also conform to MIL-P-116.

5.5.1.2 Tags and tag waterproofing. Unless otherwise specified, togs shall be waterproofed by coating both sides of the tags with a clear acrylic coating compound conforming to MIL-C-17504.

5.5.1.3 Pressure-sensitive tape application. Painted surfaces to which tape is to be applied shall be cleaned to remove dirt, grease, and other foreign matter, and then shall be coated with primer, pressure-sensltlve tape, conforming to MIL-P-3542 to increase adhesiveness of the tape and durability of seals formed by the tape.

5.5.1.4 Exposed unpainted metal surfaces.

5.5.1.4.1 Machined surfaces. Machined surfaces shall be coated with either type P-6 or P-11 preservative. Where it is necessary to protest the preservative from being prematurel$_y$ removed, such preserved areas shall be covered with grade A, greaseproof barrier-material conforming to MIL-B-121, and secured in place with type III, class 1 pressure-sensitive tape conforming to PPP-T-60. The tape and barrier-material shall be overcoated with electrical ignition waterproofing varnish conforming to MIL-V-13811.

5.5.1.4.2 Other unpainted metal surfaces. All unpainted surfaces other than those specified in 5.5.1.4.1, such as steering assemblies, tie rods, springs, pintle assemblies, threaded surfaces and surfaces exposed by disassembly, shall be coated with type P-1 preservative.

5.5.1.5 Truck. Except as otherwise specified herein, the truck and components preserved and packaged in accordance with the applicable level A requirements of MIL-STD-281.

5.5.1.5.1 Enclosed gears and gear housing such as standard drive transmissions, differentials, transfer assemblies, and power take-off assemblies and other enclosed gear trains. The gear housings not lubricated by the engine lubricating system shall be drained. After gear housings have been drained, if metal filings, metal clips, or other foreign matter is in evidence, the gear housings shall be filled with type P-2 preservative oil, which shall be agitated within housing with dry compressed air, after which the preservative oil shall be

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completely drained from the gear housings, Caution: Gears shall not be operated while the P-9 preservative oil is in the housing. The gear housings shall then be refilled to operating level with the applicable grade of gear lubricant, conforming to MIL-L-2105. If heavier grades of operting gear lubricants other than cited in above specification are recommended by the manufacturer for the equipment, such shall be used. Where gear cases require an operating oil, such cases shall be filled to operating level with type P-10, grade 2 or 3 preservative oil. A tag shall be attached indicating that the gear case must be drained and refilled with an approved operating lubricant before operation. The unit shall be operated under no load for sufficient time to ensure complete coverage of all internal surfaces within the housing, or the lubricant shall be agitated within the housing with dry compressed air of sufficient duration to ensure coverage of gears and internal surfaces of the housing. Openings into the housing shall be sealed with type III, class 1, pressure-sensitive tape conforming to PPP-T-60. The taped openings shall be sprayed with electrical ignition waterproofing varnish conformlng to MIL-V-13811.

5.5.1.5.2 Air cleaner, oil-bath type The air cleaner shall be drained and cleaned with either petroleum solvent or mineral spirits conforming to P-D-680 or TT-T-291, respectively. The air cleaner shall then be filled to operating level with type P-10, grade 2 preservative oil, and the unpainted surfaces above the operating level shall be coated with the same type and grade of preservative oil. All openings into the air cleaner shall be sealed with type III, class I pressure-sensitive tape, conforming to PPP-F-60. The taped openings shall be coated with electrical ignition waterproofing varnish conforming to MIL-V-13811.

5.5.1.5.3 Filters and strainers. Prior to service unit and truck engine preservation, filters, screens, and straomers shall be removed and inspected. Undamaged filters, screens, and strainers shall be cleaned and reinstalled. Damaged filters, screens and strainers shall be replaced with new items.

5.5.1.5.4 Engine crankcase. When applicable, the crankcase, including engine mounted accessories such us governors and other accessories that are lubricated from the engine crankcase, shall be drained. When oil sludge or other heavy contaminants are in evidence, the crankcase shall be flushed with a suitable flushing oil, or the crankcase pan shall be removed, cleaned and reinstalled. The threaded surfaces of drain plugs shall be coated with type P-1

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preservative, and the plugs reinstalled and tightened. The crankcase shall be filled to operating level with type P-10, grade 1, 2, or 3 preservative oil, as applicable. A tog containing the following information shall be attached in a conspicuous location within the driver's compartment: ENGINE CRANKCASE FILLED WITH TYPE P-10, PRESERVATIVE OIL, MIL-P-116. DRAIN CRANK-CASE AND FILL TO OPERATING LEVEL WITH OPERATIONAL LUBRICANT. VEHICLE MAY BE OPERATED UP TO 10 HOURS WITH PRESERVATIVE OIL IN CRANKCASE.

5.5.1.5.5 Diesel engine.

5.5.1.5.5.1 Preservative run-in. Using fuel conforming to VV-F-800 start the engine and run at a fast idle under no load until normal operating temperature is reached. The engine shall then be stopped and all fuel drained from the tank. The fuel tank shall be filled with type P-9 preservative oil in sufficient quantity to fiI all components of the fuel system. The return fuel line shall be disconnected from the injectors to prevent diesel fuel from diluting the preservative oil in the fuel tank. The engine shall be started and run at a fast idle under no load until it is operating on type P-9 preservative oil. The engine shall be run a sufficient time to assure that all components of the fuel system are filled with preservative oil. Then the engine shall be stopped, the retun line reconnected, and the fuel tank completely drained The preservative oil shall be allowed to remain in the fuel system. The engine shall be allowed to cool to ambient temperature and further preserved using the most applicable method that is shown under diesel engines in MIL-STD-281 . Caution: The fuel injectors or other components of the fuel system shall not be removed to apply preservative oil and the preservative oil shalll not be poured into the air intake.

5.5.1.5.5.2 Valve compartments. The cover plates shall be removed, and all residues and contaminates cleaned from internal surfaces of the com-partment, including valve mechanisms contained therein. Care shall be exercised to prevent the entry of cleaning solvents into the lubrication system. Immediately following cleaning, the engine shall be rotated with the starter and while the engine is being rotated, the rocker arm mechanism, valve sterns, springs, guides, push rods, gears and inside surfaces of cover plates shall be sprayed with type P-10, grade 2 preservative oil.

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5.5.1.5.5.3 Engine openings. Upon completion of engine preservation, all openings into the nterior of the engine, such as exhaust outlets, breathers, filler pipes, vents, and air intakes shall be sealed with type Ill, class 1 pressure-sensitive tape conforming to PPP-T-60. The tape shall be overcoated with electrical ignition waterproofing varnish conforming to MIL-V-13811.

5.5.1.5.6 Electrical equipment.

3.5.1.5.6.1 Batteries. When specified by the procuring activity (see 6.2), batteries shall be secured in place in vehicle battery carrier. The vent holes in the filler caps of dry charged batteries shall be sealed with type 1, class 1 pressure-sensitive tape conforming to PPP-T-60. The taped areas shall be overcoated with electrical ignition waterproofing varnish conforming to MIL-V-13811. These vent holes may be plugged providing such plugs will ensure positive seal against the entry of moisture. Unless otherwise specified by the procuring activity, dry charged batteries shall be packaged and packed in accordance with the overseas requirements of MIL-P-208. Electrolyte shall be packaged and pocked in accordance with requirements of MIL-S-207. Packaged batteries and electrolyte shall be stowed with other on vehicle equipment (OVE). Electrolyte shall be stowed in a manner permitting easy removal at ports when special stowing is required by maritime regulations. Battery posts shall be coated with type P-11 preservative and then covered with either plastic caps or grade C greaseproof barrier material, MIL-B-121.

5.5.1.5.6.2 Battery cables and racks. Battery cables shall be cleaned and coated with type P-11 preservative, wrapping the preserved areas with grade A greaseproof barrier material conforming to MIL-B-121. The barrier material shall be secured in place with type Ill, class 1 pressure-sensitive tape conforming to PPP-T-60. The cables shall then be secured in place on the equipment with the same kind of tape. Battery racks shall be cleaned and coated with either coating compound or asphalt varnish, conforming to MIL-C-450 and TT-V-51, respectively.

5.5.1.5.6.3 Distributors and magnetos Distribute caps and rotors shalll be removed, and all internal surfaces cleaned by wiping with a dry, clean, lint-free cloth. Using another lint-free cloth saturated with type P-9 preservative, the oil shall be applied to all metallic surfaces within the breaker compartment. When applying the preservative, care shall be taken to prevent the

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oil from coming in contact with wiring, nonmetallic ports, and contact points. The rotors and caps shall then be replaced and openings sealed with type III, class 1 pressure-sensitive tape conforming to PPP-T-60. The taped areas shall be overcoated with electrical ignition waterproofing varnish conforming to MIL-V-13811.

5.5.1.5.6.4 Generators and starters. Generators and starters shall be lubricated in accordance with the manufacturer's instructions. The drive mechanism shall be coated with type P-10, grade 2 preservative oil. If complete coverage cannot be obtained through the housing openings, the starter skill be removed and the drive mechanism coated with the above preservative oil. The starter shall then be reinstalled. The openings into generators and starters shall be sealed with type III, class 1, pressure-sensitive tape, conforming to PPP-T-60. The taped areas shall be overcooked with electrical ignition waterproofing varnish conforming to MIL-V-13811.

5.5.1.5.6.5 Wiring systems. Openings and exposed ends of plugs, sockets, terminals, voltage regulators, wiring and wiring harness shall be covered with type III, class I pressure-sensitive tape conforming to PPP-T-60, and the taped areas, including the wiring and wiring harness shall be overcooked with electrical ignition waterproofing varnish conforming to MIL-V-13811.

5.5.1.5.7 Drive belts and pulley grooves. Tension on all drive belts shall be released. Unpainted surfaces of pulley grooves shall be tooted with primer conforming to TT-P-664. A warning tag, bearing the information "BELT TENSION RELEASED: ADJUST BEFORE STARTING ENGINE," shall be securely attached in a conspicuous location within driver's compartment to the compressor clutch control handle. As an alternative, drive belts may be removed and packaged either individually for single drive or together in sets for multiple drives, in accordance with method IC-3. If drive belts are removed, a tog shall be attached to the unit indicating that drive belts are to be installed prior to engine operation. The packaged belts shall be placed in the tool box.

5.5.1.5.8 Vehicle cab and body. In modification of requirements specified in MIL-STD-281, the windshield wiper arm and blades shall be removed and, together with keys, shall be packaged in accordance with method IC-3 and then stowed in dash comportment, or securely attached to steering column. Neither the cover nor the top bows shall be removed from the vehicle body.

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5.5.1.5.9 Underside of vehicles. The exposed surfaces of the frames, underside of running boards, floor panels, fenders and fuel tanks shall be coated with under body coating, conforming to TT-C-520. Core shall be exercised to keep the material off of such components as springs, brake lines, tie rods, steering mechanisms, etc.

5.5.1.5.10 Fire extinguishers. All openings into the interim of the fire extinguishers shall be sealed with type III, class 1, pressure-sensitive tape conforming to PPP-T-60. All exterior surfaces including the taped areas shall be overcoated with a clear acrylic coating compound conforming to MIL-C-17504. The extinguisher shall then be packaged in accordance with method IC-5 and pocked with other OVE.

5.5.1.5.11 Lamps and reflector lenses. For rail shipment only, exterior lamp and reflector lenses (red or amber) shall be completely covered with wrapping paper, conforming to MIL-P-130. The wrapping paper shall be covered and secured with tape conforming to type III, class 1 of PPP-T-6O.

5.5.1.5.12 Personnel heater. If truck is equipped with a personnel heater, then exterior metal surfaces of the heater subject to corrosion shall be coated with type P-1 preservative compound. Openings into the heater shall be covered and electric motors shall be wrapped or covered with grade C, greaseproof barrier-material conforming to MIL-B-121, secured in place with type III, class 1 pressure-sensitive tape conforming to PPP-T-60. The tape and barrier-material shall be overcoated with electrical ignition waterproofing varnish, conforming to MIL-V-13811.

5.5.1.5.13 Exterior surfaces of engine and interior surfaces of engine Compartment. The entire surfaces of the engine, and engine-mounted accessories, including taped openings, sealed connections, electric wiring, wiring harness, and interior surfaces of the engine compartment and hood shall be coated with electrical ignition water-proofing varnish conforming to MIL-V-13811.

5.5.1.6 Service unit.

5.5.1.6.1 Hydraulic system. The supply tank of the hydraulic system on the service unit and the coolant system shall be checkedd and filled to operating level with an approved hydraulic fluid and coolant as recommended by the manufacturer.

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5.5.1.6.2 Controls. The controls of the service unit shall be placed in the following positions:

(a) Fuel valves - intake positions.

(b) Fan motor and fuel pump - off position.

(c) Agitator - off position.

(d) Compressor clutch control handle - disengaged position.

(e) All air valve controls slightly open except safety air valve, which shall be placed in a completely open position.

(f) Coolant valves - closed position.

(g) Hydraulic tank drain valve - closed position.

(h) Compressor oil drain valve - closed position. No processing required.

(i) Cooler drain valve - open position.

5.5.1.6.3 Control panel. The exposed high pressure lines on the back of the control panels shall be protected from damage during shipment and storage with either a plywood or sheet metal cover secured to the control panel housing.

5.5.1.6.4 Grounding reel assembly. The openings into the reel drum shall be sealed off with type III, class 1 pressure-sensitive tape conforming to PPP-T-60. The exposed portions of the grounding cable and clip shall be covered with the above cited tape. The loose end of the taped grounding cable shall be secured to the truck body with the same pressure-sensitive tape.

5.5.1.6.5 Gages and instruments. The glass or plastic dials of gages and instruments on the control console of the service unit shall be covered with o fitted piece of weather-resistant solid fiberboard or plywood, conforming to PPP-B-636 or NN-P-515, secured in place with type Ill, class 1 pressure-sensitive tape conforming to PPP-T-60. The tape, fiberboard or plywood and exposed metal surfaces of gages and instruments shall be coated with electrical ignition waterproofing varnish conforming to MIL-V-13811.

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5.5.1.6.6 Hoses and hose fittings. All hose fittings shall be wrapped with type 1 laminated and creped wrapping paper conforming to MIL-P-130, or with grade A greaseproof barrier material conforming to MIL-B-121. Wraps shall be secured in place by tying or taping. All hoses shall be neatly coiled in the hose and tool box. The hose and tool box lid retainer that has one end attached to the service unit mixing tank shall have the base end of the retainer (chain) secured to the tank body with the same kind of tape as cited in 5.5.1.6.5.

5.5.1.6.7 High pressure air service valves, bleed valves, and high pressure air service connections. These valves and connections, which are located in the same areas on the outside of truck body, shall be protected from damage in the same area on the outside of truck body, shall be protected from damage during shipment and storage with a sheet metal cover securely affixed to the truck-body by any means not requiring the drilling of holes in truck body

5.5.1.6.8 Fuel mixing system. The entire fuel mixing system shell be flushed out with type P-10, grades 2 or 3 preservative oil, allowing excess preservative oil to drain off. The interim surfaces of the mixing tank and cover shall be tooted with the same type of preservative oil.

5.5.1.6.9 Fuel service outlet and inlet. Each fuel service inlet adn outlet shall be closed with the cap or plug provifefd and wrapped with type 1 laminated and creped wrapping paper conforming to MIL-P-130 or with grade A grease-proofed barrier material conforming to MIL-B-121. The wraps shall be secured in place by tying or taping.

5.5.1.6.10 Suction pipe and stowage tube. The suction pipe shall be coated with type P-9 preservative and placed in the storage tube. The stowage tube cover shall be closed and the seams around cover taped with type Ill, class 1, pressure-sensitive tape conforming to PPP-T-60. The tape shall be overcoated with electrical ignition waterproofing varnish conforming to MIL-V-13811.

5.5.1.6.11 Air compressor. Air compressor shall be preserved and packaged in accordance with level A requirements of MIL-C-51172.

5.5.1.7 Nameplates, data and instruction plates. These items shall be thorougly cleaned, dried and then coated with a clear acrylic coating compound conforming to MIL-C-17504.

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5.5.1.8 Components and repair parts. Components and repair parts accompanying equipment shall be preserved and packaged in accordance with MIL-R-196 and MIL-P-116.

5.5.1.9 Tools. Tools accompanying equipment shall be preserved and packaged in accordance with MIL-H-15424.

5.5.1.10 Technical Publications. Technical publications shall be packaged in accordance with method IC-3.

5.5.1.11 Intermediate packaging.

5.5.1.11.1 Components and accessories. Components and accessories not attached to or disassembled from the basic unit (see 5.1) preserved and packaged as specified in 5.5.1.5 and 5.5.1.8 shall be intermediately packaged in snug-fitting overseas type containers conforming to PPP-B-621 or to PPP-B-601. The gross weight of each cantainer shall not exceed 150 pounds. Internal cushioning and blocking shall be used as required to make a tight package. The containers shall be closed and strapped in accordance with the applicable container specification and the appendix thereto, using galvanized strapping, of after which the containers shall be secured in a protected location on the equipment,

5.5.1.11.2 Repair parts, tools, and technical publications. Repair ports, tools, and technical publications preserved and packaged as specified in 5.5.1.8 through 5.5.1.10 shall be placed in the tool chest provided. Cushioning shall be used as required to prevent movement within chest. The chest cover hinges and latches shall be oiled with P-7 preservative oil, and excess oil shall be wiped off. The chest cover shall then be closed and secured, and all seams around cover shall be taped with type III, class 1 pressure-sensitive tape conforming to PPP-T-60. The tape shall be overcoated with electrical ignition waterproofing varnish conforming to MIL-V-13811.

5.5.2 Level C. The truck, service unit apparatus, including components, accessories, repair parts, tools and technical publications, cleaned and dried as specified in 5.2 shall be preserved and packaged in accordance with level C requirements of MIL-STD-281.

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5.6 Packing, levels A, B or C. The service unit, including components, accessories, repair ports, tools and technical publications preserved and packaged as specified in 5.5 shall be shipped mobile in accordance with the applicable level A, B or C requirement of MIL-STD-281.

5.7 Marking. In addition to any special marking required by the contract or order, unit packages, shipping containers, and the vehicles shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This unit is intended for use in servicing both portable and mechanized flame throwers with thickened fuel and compressed air.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level or levels of preservation, packaging, and packing.

(c) Whether dry batteries are to be secured in place in the vehicle battery carrier (see 5.5.1.5.6.1).

(d) Additional identification plates required.

(e) Serial number of each compressor and serial number of each service unit.

(f) Whether Drawing DL81-8-751 or DL81-8-1049 is required.

6.3 Retesting for stress. Any internal pressure applied after heat treatment and previous to the official test should not exceed 90 percent of the test pressure.

6.4 Test apparatus failure. If the test apparatus fails before the pressure has been maintained for a period of time sufficient to determine the permanent volumetric expansion, the test may be repeated at a pressure increase of 100 psig when the air receiver is tested.

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6.5 Caution. Because of the hazards inherent in testing with high pressures, manufacturers are hereby notified that every precaution Possible should be token for the safety and protection of personnel engaged in testing the items.

6.6 Repair parts packaging. When repair parts packaging is required for items authorized for stocking as repair parts, the repair parts packaging data sheets covering the applicable components of the service unit should be used. (Copies of repair parts packaging data sheets may be obtained from the Director of Engineering & Industrial Services, ATTN: Chief, Design Branch, US Army Edgewood Arsenal, Edgewood Arsenal, Maryland - 21010.)

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)

Project No. 1040-A033

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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